PYTHON & DATA SCIENCE ROADMAP

33 Structured Practice Sessions | 60–90 Minutes Per Topic

Session Execution Standard

Paste **one numbered topic** into your prompt placeholder per 60–90 minute session. Do not combine topics within a single session. Maintain strict progression across all 33 topics.

PART A: PYTHON FUNDAMENTALS (SESSIONS 1–15)

#	TOPIC & SCOPE	SESSION TIME
1	**Variables & Operations:** Data types, type conversion, input/output, arithmetic & logical operators	60–90 min
2	**Strings In-Depth:** Indexing, slicing, string methods (`split`, `join`, `strip`, `replace`), f-strings formatting	60–90 min
3	**Control Flow:** Conditionals (`if / elif / else`), comparison operators, complex logical statements	60–90 min
4	**Iteration & Loops:** `for`, `while`, `range()`, loop control (`break / continue`), nested loops, pattern printing	60–90 min
5	**Lists & Sequence Data:** Indexing, slicing, list methods, custom sorting, multidimensional/nested lists	60–90 min
6	**Tuples & Sets:** Immutability vs mutability, uniqueness guarantees, set operations, performance trade-offs	60–90 min
7	**Dictionaries & Hash Maps:** Frequency counting, grouping data, nested dictionaries, handling lists of dictionaries	60–90 min
8	**Functions & Modularization:** Positional/keyword parameters, return values, default arguments, variable scope	60–90 min
9	**Functional Features:** `*args`, `**kwargs`, lambda expressions, `map`, `filter`, key-based `sorted`	60–90 min
10	**Comprehensions:** List comprehensions, dictionary comprehensions, conditional filtering inside expressions	60–90 min
11	**Iterators & Generators:** Iteration protocol, custom generators, memory efficiency using `yield`	60–90 min
12	**Decorators:** First-class functions, wrapper functions, syntax sugar (`@decorator`), passing arguments	60–90 min
13	**File I/O Handling:** Text files, reading/writing CSV files, parsing JSON data structures, context managers	60–90 min
14	**Exception Management:** `try / except / else / finally`, custom exception classes, traceback debugging	60–90 min
15	**Modules & Standard Library:** Code importation, utilizing `collections`, `datetime`, `random`, `os`	60–90 min

PART B: OBJECT-ORIENTED PROGRAMMING (SESSIONS 16–21)

#	TOPIC & SCOPE	SESSION TIME
16	**Classes & Objects:** Instantiation, `__init__` constructor, instance attributes, methods, `self` binding	60–90 min
17	**Class Architecture:** Dunder methods (`__str__`, `__repr__`), class vs instance attributes, `@classmethod`, `@staticmethod`	60–90 min

PART C: SCIENTIFIC COMPUTING WITH NUMPY (SESSIONS 22–26)

#	TOPIC & SCOPE	SESSION TIME
22	**Array Fundamentals:** N-dimensional arrays, data types (`dtype`), array shapes, reshaping, memory layout	60–90 min
23	**Indexing & Selection:** Multi-dimensional slicing, fancy indexing, boolean masking for data filtering	60–90 min
24	**Vectorization & Broadcasting:** Universal functions (ufuncs), broadcasting rules, axis-wise aggregations	60–90 min
25	**Utility Functions:** np`.where`, sorting, `unique`, array concatenation, random generation, basic linear algebra	60–90 min
26	**NumPy Optimization Challenge:** Refactoring slow Python loop patterns into vectorized NumPy computations	60–90 min

PART D: DATA ANALYSIS WITH PANDAS (SESSIONS 27–33)

#	TOPIC & SCOPE	SESSION TIME
27	**Data Structures:** Series & DataFrames, loading CSV/JSON files, initial inspection (`head`, `info`, `describe`)	60–90 min
28	**Data Selection & Subsetting:** Explicit indexing with `loc` and `iloc`, conditional filtering, column operations	60–90 min
29	**Data Cleaning:** Missing value treatment (`isna`, `fillna`, `dropna`), deduplication, type casting, renaming	60–90 min
30	**Transformations:** Applying functions (`apply`, `map`), creating derived columns, string & datetime accessor methods	60–90 min
31	**Aggregation & Summarization:** Split-apply-combine with `groupby`, custom aggregations (`agg`), pivot tables, `value_counts`	60–90 min
32	**Combining Datasets:** Relational joins with `merge`, index-based `join`, structural concatenation (`concat`)	60–90 min
33	**End-to-End Capstone:** Analyze real-world Kaggle dataset (e.g., Titanic) to answer 15 targeted analytical questions	60–90 min

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Session Input Format:
Today's topic: 7. Dictionaries: counting, grouping, nested dicts, a list of dicts
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Execution System

- **Sequential Strictness:** Complete 1–15 first; foundational concepts are non-negotiable prerequisites.
- **Fast-Track Option:** If experienced, fast-track topics 1–4 and start intensively at topics 5 & 7.
- **Weekly Stress Test:** Every Sunday, run a practice exam covering all topics completed during the week.
- **Milestone Review:** Conduct a comprehensive mixed-topic exam across all 33 units prior to evaluation dates.

How to Master Coding Problems

1. **Deconstruct First:** Trace input/output examples manually on paper before writing code.
2. **Isolate Logic:** Break down complex problems into plain English pseudo-code steps.
3. **Verify Constraints:** Test edge cases (empty inputs, zeroes, negative values) systematically.